Implementation of Subthreshold Adiabatic Logic for Ultralow-Power Application

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***Abstract*— Behavior of adiabatic logic circuits in weak inversion or subthreshold regime is analyzed in depth for the first time in the literature to make great improvement in ultralow-power circuit design. This novel approach is efficacious in low-speed operations where power consumption and longevity are the pivotal concerns instead of performance. The schematic and layout of a 4-bit carry look ahead adder (CLA) has been implemented to show the workability of the proposed logic. The effect of temperature and process parameter variations on subthreshold adiabatic logic-based 4-bit CLA has also been addressed separately. Postlayout simulations show that subthreshold adiabatic units can save significant energy compared with a logically equivalent static CMOS implementation. Results are validated through extensive simulations in 22-nm CMOS technology using CADENCE SPICE Spectra.**

***Index Terms*— Adiabatic logic, carry look ahead adder (CLA), leakage, low power, subthreshold.**

I. Introduction

THE demand for implementing ultralow-power digital systems in many modern applications, such as mobile systems, sensor networks, and implanted biomedical systems, has increased the importance of designing logic circuits in subthreshold regime. These emerging applications have low energy as the primary concern instead of performance, with the eventual goal of harvesting energy from the environment. In subthreshold logic [1]–[4], circuits operate with a supply voltage V_{DD} lower than the transistor threshold voltage V_T and utilize the subthreshold leakage current as the operating current. Conventional CMOS logic circuits utilizing subthreshold transistors can typically operate with a very low power consumption, which is mainly due to the dynamic (switching) power consumption and is quadratically dependent upon the supply voltage as $C_L f V_{\mathrm{DD}}^2$ (where C_L, f, and V_{DD} are the load capacitance, operating frequency, and the supply voltage, respectively).

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Recently, adiabatic logic (or energy recovery logic) [5]–[8] style has emerged as a promising approach in strong inversion regime, to reduce dynamic power consumption significantly without sacrificing noise immunity and driving ability. These circuits achieve ultralow energy consumption by steering currents across devices with low voltage differences and by gradually recycling the energy stored in their capacitive loads, especially in low-frequency regime. Since the performance requirements are quite relaxed in many of these energy-efficient subthreshold applications [1], [2], we believe that the adiabatic style can be used efficaciously in a subthreshold regime to make the circuit more energy efficient. To the best of our knowledge, no paper emphasizes the application of adiabatic logic in weak inversion regime for advanced technology node such as 22 nm. Therefore, the attempt to realize the subthreshold adiabatic logic (SAL) concept is a new endeavor.

In general, the design of SAL requires a deep knowledge of the main features of the adopted logic style, such as power dissipation, leakage current, impact of temperature variation, operating frequency, and noise immunity. In this paper, the behaviors of adiabatic logic in subthreshold regime are discussed in depth. To demonstrate the workability of the adiabatic logic circuits in subthreshold regime, a 4-bit carry look ahead adder (CLA) unit is adopted as a reference circuit. Analysis of energy dissipation along with the sensitivity of energy dissipation on supply voltage and temperature variations is also discussed in depth. Moreover, the analytical expression of optimum frequency and supply voltage under minimum energy condition have been verified through simulation in 22-nm technology. Extensive experiments are carried out using CADENCE SPICE Spectra to ensure the high energy efficiency and design feasibility of the proposed logic in weak inversion regime compared with other conventional CMOS logic.

The rest of this paper is organized as follows. Section II explains basic behavior of the MOS transistor under adiabatic switching in weak inversion regime. An analysis of power dissipation, leakage current, temperature sensitivity, optimum frequency, and supply voltage, etc., of proposed subthreshold adiabatic logic circuits are detailed in Section III. The efficacy of adiabatic approach over conventional CMOS is also detailed in terms of experimental results in Section III. The conclusion is given in Section IV.

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II. Subthreshold Adiabatic Logic

A. Basic Model of MOSFET in Subthreshold Regime

Using EKV model [9], the I–V characteristics of the subthreshold pMOS device can be expressed by

$$I_{\text{Sub}} = I_0 e^{\frac{(V_{\text{SG}}-|V_{\text{TH}}|)}{n_p V_T}}\left(1 - e^{\frac{-V_{\text{SD}}}{V_T}}\right) \tag{1}$$

where $I_0 = 2\eta_p \mu_p C_{\text{ox}}(W/L)V_T^2$. V_{SG}, V_{SD}, and V_{TH} are source to gate, source to drain, and threshold voltage of pMOS, respectively. $V_T = (kT/q)$ is thermal voltage, η_p is subthreshold slope factor, and μ_p is the mobility of pMOS device. In subthreshold regime, the threshold voltage (V_{TH}) depends on source-to-drain voltage (V_{SD}) through body effect and drain-induced barrier lowering. Considering these effects, threshold voltage can be expressed as follows:

$$V_{\text{TH}} = V_{\text{TO}} - \gamma V_{\text{SB}} - \eta V_{\text{SD}}. \tag{2}$$

Here, V_{TO} is the threshold voltage at zero bias and γ and η are the body effect and drain-induced barrier lowering coefficient, respectively. The Drain Induced Barrier Lowering coefficient (η) [9], [10], which is a unitless quantity, can be expressed as

$$\eta = \frac{1}{2\cosh\left(L_{\text{eff}}/2l_t\right)} \tag{3}$$

where $l_t = (\varepsilon_{\text{Si}} T_{\text{ox}} X_{\text{Dep}}/\beta\varepsilon_{\text{ox}})^{1/2}$ and

$$\gamma = \sqrt{\frac{2q\varepsilon_{\text{Si}} N_A}{C_{\text{ox}}}}. \tag{4}$$

For conventional CMOS configuration (source and body are tied together), (1) can be expressed as

$$I_{\text{SUB}} = K_P \exp\left(\frac{V_{\text{SG}}}{n_P V_T}\right)\left(1 - \exp\left(\frac{-V_{\text{SD}}}{V_T}\right)\right) \tag{5}$$

where $K_P = I_0 \exp(-V_{\text{TO}} - \eta V_{\text{SD}}/n_P V_T)$. For the sake of simplicity, let us further simplify the magnitude of K_p by assuming, $\eta V_{\text{SD}}/n_p V_T \ll 1$. The value of η can be obtained directly from the 22-nm PTM model file. Putting the model parameters in (3) and (4), η will be slightly less, approximately 0.0039. Hence, we set, $\eta \to 0.004$ for pMOS and nMOS both, thermal voltage, $V_T = 26$ mV at 27 °C, $n_p = 1.62$, V_{SD} lies in the range of few millivolts to hundreds of millivolts depending on the V_{SG} of the pMOS device. In the case of a pMOS device, we get $K_P \approx I_0 \exp(-V_{\text{TO}}/n_P V_T)$. K_p and K_n differ due to imbalance in zero biased threshold voltage (V_{TO}), mobility (μ), subthreshold swing (n), and so on. In 22-nm technology, as I_0 for pMOS and nMOS are 1.16 and 0.55 μA, respectively, K_p and K_n will be obtained by putting the value from PTM 22-nm metal gate/high-K model file [11]. Hence, $K_P \sim 1.25$ and $K_n \sim 8.79$ pA. By changing (W/L) ratio, we can also tune K_p or K_n as these are directly proportional to the aspect ratio of transistors.

B. Behavior of MOSFET Under ac Supply Voltage in Subthreshold Regime

In adiabatic logic circuits, ramp type supply voltage is used to slow down the charge transport mechanism [5], [6].

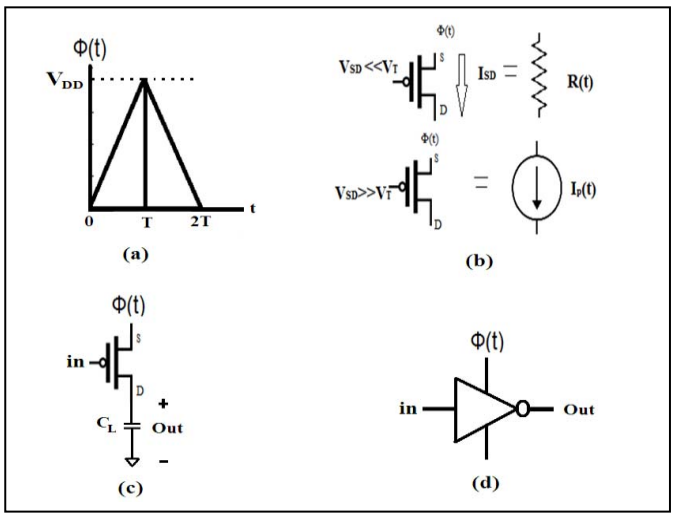


Fig. 1. (a) Clock supply voltage. (b) Equivalent circuit of pMOS under low and high V_{SD}. (c) SAL inverter circuit. (d) SAL inverter symbol.

Hence, the supply clock plays the pivotal role. A ramp type supply voltage $\phi(t)$ is considered in Fig. 1(a), which gradually swings in between logic 0 (Gnd potential) and logic 1 (V_{DD}) in time duration $2T$, where f (=1/2T) is the supply clock's frequency. The power supply waveform $\phi(t)$ can be divided into charging phase, when $\phi(t)$ ramps up from 0 to V_{DD} in 0 to T unit time and discharging phase when $\phi(t)$ ramps down from V_{DD} to 0 in T to $2T$ unit time. Considering the charging phase, the voltage at any time instant can be expressed as

$$\phi(t) = V_{\text{DD}}\left(\frac{t}{T}\right). \tag{6}$$

Hence, the behavior of pMOS under high and low V_{SD} will be examined. When we apply logic 1 at the gate of pMOS, V_{SD} becomes high and the drain current can be rewritten as

$$I_{\text{SUB},P} = K_P \exp\left\{\frac{V_{\text{DD}}}{n_P V_T}\left(\frac{t}{T} - 1\right)\right\} \tag{7}$$

$(1-\exp(-V_{\text{SD}}/V_T)) \sim 1$ when $V_{\text{SD}} \gg V_T$. Similarly, when we apply logic 0 at the gate of pMOS, very small voltage drops across the transistor. Under this condition, we get

$$I_{\text{SUB},P} = K_P \exp\left(\frac{V_{\text{DD}} t}{n_P V_T T}\right)\left(\frac{\Delta V}{V_T}\right) \tag{8}$$

where

$$\Delta V = \phi(t) - V_{\text{OUT}} = C_L \frac{\partial V_{\text{OUT}}}{\partial T} \tag{9}$$

where ΔV and V_{OUT} are the voltage drops across the resistor and the output node voltage, respectively. Thus, when looking at the source and the drain terminals under low V_{SD}, the pMOS transistor is equivalent to time-varying resistor

$$R_p(t) = \frac{V_T}{K_P}\exp\left(-\frac{V_{\text{DD}} t}{n_P V_T T}\right). \tag{10}$$

Equivalent circuit of pMOS under high and low V_{SD} are given in Fig. 1(b). Indeed, ΔV is a time-dependent voltage drop.

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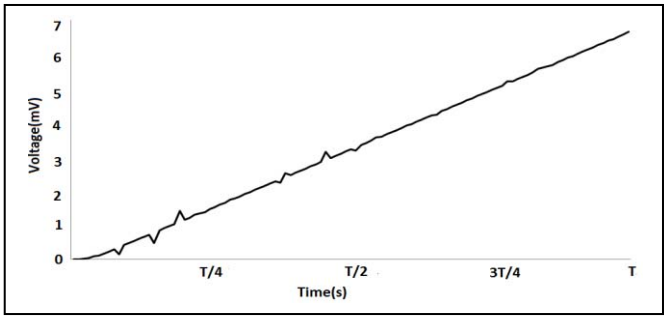


Fig. 2. Variation of ΔV with respect to time.

Solving (9), we obtain

$$\Delta V = n_P V_T \exp\left\{-\frac{\beta}{\alpha}\left(\exp\left(\frac{\alpha t}{T}\right)\right)\right\} \times\left[E_i\left\{\frac{\beta}{\alpha}\left(\exp\left(\frac{\alpha t}{T}\right)\right)\right\} - E_i\left(\frac{\beta}{\alpha}\right)\right]. \quad (11)$$

Here, $\beta = n_P T/C_L R_p(t)$; $\alpha = V_{\rm DD}/n_P V_T$; $C_L R_p(t)$ is the product of the path resistance and output load capacitances under low $V_{\rm SD}$; $E_i(x)$ is the exponential Integral [12] of x, where $E_i(x) = \int \exp(x)/x$. Putting $\gamma = \beta/\alpha \exp(\alpha t/T)$, the above equation can be rewritten as

$$\Delta V = n_P V_T \exp(-\gamma)\left[E_i(\gamma) - E_i\left(\frac{\beta}{\alpha}\right)\right]. \quad (12)$$

The time dependency of the voltage drop (ΔV) across the resistor is given in Fig. 2. Approximately ΔV can be treated as constant over the entire pulsewidth due to the negligible voltage variation with time. Under this assumption, energy dissipation can be expressed as

$$C_L \frac{\partial V_{\rm OUT}}{\partial T} = I_{\rm SUB} = \frac{K_P}{V_T} \exp\left(\frac{V_{\rm DD} t}{n_P V_T T}\right)\Delta V. \quad (13)$$

This gives

$$\Delta V = V_{\rm DD}\left(\frac{V_T K_P}{C_L T}\right) \quad \text{when } 0 < t \le T. \quad (14)$$

An SAL inverter circuit and the symbols are given in Fig. 1(c) and (d). In the case of an SAL inverter, if in $= 0$, the pMOS transistor will behave as a resistor due to low voltage drop across the source and the drain terminals of the transistors. When $\phi(t)$ ramps up from 0 to $V_{\rm DD}$, the output capacitor (C_L) will be charged up to $V_{\rm OH}$. When supply clock falls down from $V_{\rm DD}$ to 0, the charge stored in the output capacitor will be sent back to the supply clock. Hence, at $t = T/2$, $V_{\rm OH}$ can be expressed as

$$V_{\rm OH} = V_{\rm DD} - \Delta V \quad (15)$$

where ΔV is given in (11). When we apply in $= 1$, the pMOS transistor behaves as a current source; voltage drop across the output rises up to $V_{\rm OL}$. Hence, as the supply voltage ramps up and down, leakage current will not be constant also. Maximum leakage current will flow close to $t = T/2$. Hence, $V_{\rm OL,}$ due to leakage flow can be expressed as follows:

$$V_{\rm OL} = \exp(-0.5\alpha)\frac{T n_P K_P V_T}{C_L V_{\rm DD}}\{1 - \exp(-0.5\alpha)\}. \quad (16)$$

Hence, $V_{\rm OL}$ is negligibly small as $\alpha \gg 1$ or $V_{\rm DD}/n_p V_T \gg 1$.

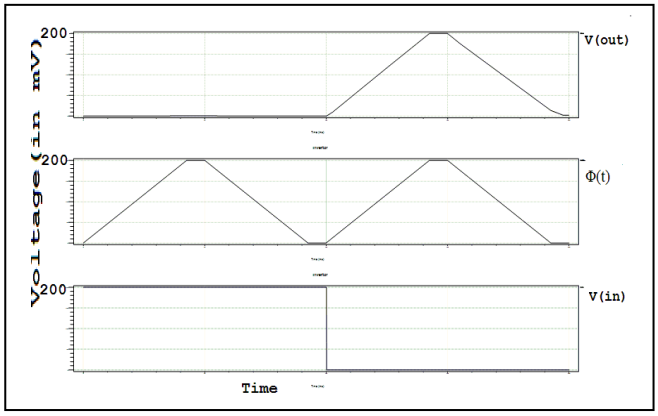


Fig. 3. Input output waveforms of an SAL inverter. $\Phi(t)$ is a time-varying supply voltage. Here, $V_{\rm in}$ and $V_{\rm out}$ are the input and output voltages of the SAL inverter given in Fig. 1(c).

The energy dissipation ($E_{\rm ch,ad}$) in the above-mentioned SAL inverter during the charging process can be calculated as

$$E_{\rm ch,ad} = \int_0^T \frac{(\Delta V)^2}{R} dt = \frac{n_P V_{\rm DD}}{T}\frac{V_T^2}{K_P} C_L^2\left(1 - \exp\left(-\frac{V_{\rm DD}}{n_P V_T}\right)\right). \quad (17)$$

Since $V_{\rm DD}/n_p V_T \gg 1$, the above equation becomes

$$E_{\rm ch,ad} = \frac{1}{T}\left(\frac{C_L V_T}{K_P}\right) C_L V_{\rm DD}^2. \quad (18)$$

The input and output waveforms of the SAL inverter circuit along with the supply clock are given in Fig. 3.

Considering the charging and discharging phases, the total power dissipation due to switching can be expressed as

$$E_{\rm switch,ad} = \frac{2}{T}\left(\frac{C_L V_T}{K_P}\right) C_L V_{\rm DD}^2. \quad (19)$$

For conventional subthreshold operation, using a constant dc power supply and the same input condition, we can estimate the total energy dissipation ($E_{\rm switch,conv}$) across a pMOS transistor over $0 < t \le T$ as follows:

$$E_{\rm switch,conv} = \int_0^T \frac{(\Delta V)^2}{R} dt = \frac{n_P V_T}{K_P}\frac{C_L^2 V_{\rm DD}^2}{T}. \quad (20)$$

Considering Gain(%) $\{= E_{\rm switch,ad}/E_{\rm switch,conv} \times 100\}$, it could be stated that a pMOS using subthreshold adiabatic logic consumes only 43.47% of the total energy consumed by the static CMOS counterpart. For different frequencies, variation of the gain in the case of an SAL inverter is given in Fig. 4. From Fig. 4, we can see that the minimum gain is 41% at nearly 200 KHz and the maximum gain is almost 48% at nearly 50 KHz. It can also be seen that gain varies from 44% to 42%, which confirms the theoretical calculations of energy gain (43.43%) as shown in the range of 100–500 KHz. As the frequency decreases, the gain increases but the voltage level degrades at the output node. In a low frequency range of the order of few hertz, we obtain almost 70% and 23% of supply voltage as output high ($V_{\rm OH}$) and low voltage ($V_{\rm OL}$), respectively.

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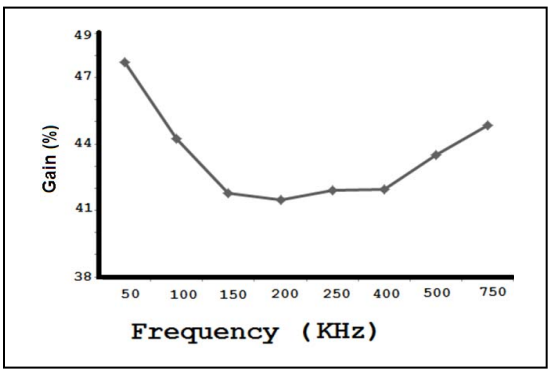


Fig. 4. Gain in terms of switching energy dissipation for varying frequencies.

C. Impact of Parasitic Gate Capacitances on Variable Supply Voltage

In a subthreshold digital CMOS circuit, the gate capacitance of the transistor varies if the supply voltage ramps up (charging phase) or down (discharging phase) [13], [14]. Due to this, a negligible amount of energy is transferred from variable supply voltage to the inputs, causing a nonadiabatic loss. This nonadiabatic loss of energy also brings noise or some unwanted spikes in output node voltages. Since during adiabatic switching, MOSFETs are kept under low drain to source voltages, the total gate capacitance (C_G) of a pMOS device can be expressed as

$$C_G = C_{\rm GD} + C_{\rm GS}$$

where $C_{\rm GD}$ and $C_{\rm GS}$ are the gate-to-drain and gate-to-source capacitances, respectively. It can be assumed that initially the gate voltage and supply voltages are at zero potential. When the supply voltage ramps up, within a short time span, the gate voltage rises gradually from 0 to $V_{\rm DD}$. As a result, the total $C_G V_{\rm DD}$ amount of charge flows to the gate node, causing a 0.5 $C_G(V_{\rm DD})^2$ J of energy loss. When the supply voltage reaches the peak of $V_{\rm DD}$, the drain-to-gate voltage ($V_{\rm DG}$) and the source-to-gate voltage ($V_{\rm SG}$) become zero, causing a current flow from the gate node to the input line. The energy associated during this event is $C_G V_{\rm DD}^2$. As a result, a 0.5 $C_G(V_{\rm DD})^2$ energy flows from the gate node to the input.

D. Impact of Leakage for Variable Supply Voltage

As leakage current is one of the pivotal concerns in subthreshold regime [14]–[17], total leakage energy dissipations in SAL along with the conventional one are given. Hence, leakage current is assumed to flow through the pMOS device for higher drain to source voltage drop. Under this condition, for the half clock period of supply voltage, φ (t), the total leakage energy dissipation across a pMOS can be expressed as

$$E_{\rm Leak,ad} = \int_0^{T/2} K_P \exp\left\{\frac{V_{\rm DD}}{n_P V_T}\left(\frac{t}{T}-1\right)\right\}\phi(t)dt. \quad (21)$$

Assuming $V_{\rm DD}/n_P V_T \gg 1$, we get

$$E_{\rm Leak,ad} \approx \exp\left(-\frac{V_{\rm DD}}{n_P V_T}\right)\left\{\frac{T K_P n_P V_T}{2}\left(1-\frac{2n_P V_T}{V_{\rm DD}}\right)\right\}. \quad (22)$$

Considering the constant dc supply voltage having the same value as the peak of $\varphi(t)$, the total leakage across a pMOS device can be expressed as

$$E_{\rm Leak,conv} = V_{\rm DD} T K_P. \quad (23)$$

Comparing (22) and (23), we get

$$\frac{E_{\rm Leak,conv}}{E_{\rm Leak,ad}} = \frac{2\eta^2 e^\eta}{\eta - 1} \approx 2\eta e^\eta \quad (24)$$

where

$$\eta = \frac{V_{\rm DD}}{2n_P V_T} \gg 1.$$

It can be observed that the conventional one having a constant dc supply consumes $2\eta e^\eta$ times more leakage energy compared with SAL.

E. Optimum Frequency and Supply Voltage Calculation of SAL Inverter

The total energy for the entire time duration T in an SAL inverter is given by

$$E_{\rm Total,ad} = \frac{2}{T}\left(\frac{C_L V_T}{K_P}\right) C_L V_{\rm DD}^2 + E_{\rm leakage}. \quad (25)$$

The first term corresponds to the switching energy dissipation and $E_{\rm leakage}$ is the leakage power dissipation in the SAL, which can be estimated from (22). Therefore, (25) can be rewritten as

$$E_{\rm Total,ad} \approx \frac{2}{T}\left(\frac{C_L V_T}{K_P}\right) C_L V_{\rm DD}^2 + \frac{2T K_P n_p^2 V_T^2}{V_{\rm DD}} \times \left\{1-\exp\left(-\frac{V_{\rm DD}}{2n_p V_T}\right)\left(1+\frac{V_{\rm DD}}{2n_p V_T}\right)\right\}. \quad (26)$$

Differentiating (26) and putting $\partial E_{\rm Total,ad}/\partial T = 0$, we get

$$f_{\rm opt} \approx \sqrt{\frac{K_P^2 n_p^2 V_T}{2C_L^2 V_{\rm DD}^3}\left\{1-\exp\left(-\frac{V_{\rm DD}}{2n_p V_T}\right)\left(1+\frac{V_{\rm DD}}{2n_p V_T}\right)\right\}}. \quad (27)$$

Hence, $f_{\rm opt}$ is the optimum frequency at which the SAL inverter consumes minimum energy. Simulated and theoretical values for optimum frequency in the case of an SAL inverter are given in Fig. 5. In 22 nm, the theoretical optimum frequency of 51.3 KHz is very close to the simulated frequency of 51.25 KHz.

Similarly, we can find out the optimum supply voltage by putting $\partial E_{\rm Total,ad}/\partial V_{\rm DD} = 0$. Therefore, optimum $V_{\rm DD}$ for an SAL inverter can be estimated as

$$V_{\rm DD,opt} \approx \sqrt[3]{\frac{K_P K_N n_n^2 T^2 V_T}{2C_L^2}}. \quad (28)$$

Theoretical and simulated values are shown in Fig. 6 to justify the accuracy of the analytical expression of optimum $V_{\rm DD}$. Hence, the theoretical optimum supply voltage of 190.77 mV is very close to 190 mV, which is the simulated value of optimum supply voltage.

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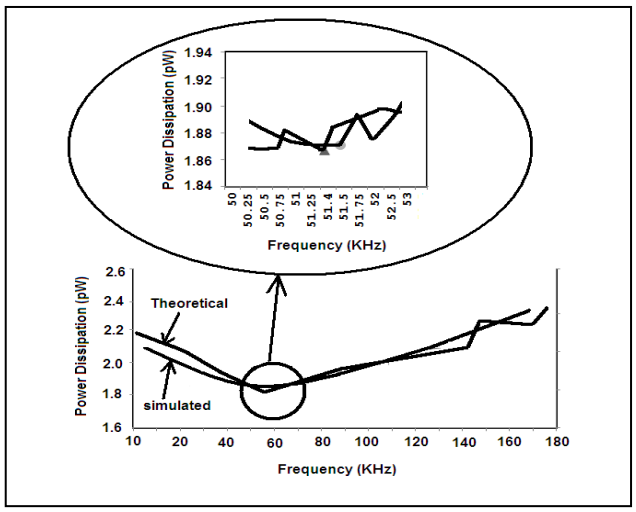


Fig. 5. Power dissipation of an SAL-based inverter for different frequencies.

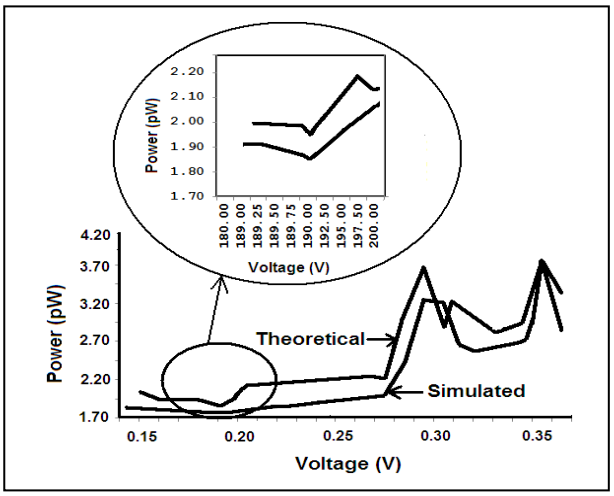


Fig. 6. Power dissipation of an SAL-based inverter for different supply voltages.

F. Delay Calculation

As SAL is efficacious where instead of performance, power dissipation is major concern. For example, in implanted biomedical systems, the circuits remain active for a very small span of time and remain idle for most of the time. In such operations, much lower frequency ranges are required. Therefore, in SAL, minimization of power dissipation is the pivotal issue. Hence, we address the delay in SAL though it would be few times larger than that in the conventional one. In SAL-based digital circuits, output nodes follow the supply clock very closely during the charging and discharging periods and the output waveforms get the same pattern as the supply voltage. In general, delay can be calculated between a change in an input (50% of V_{DD} on input) and a low-to-high or high-to-low change in the output (50% of V_{DD} on input). As the supply voltage ramps up and down linearly in between 0 and V_{DD}, propagation delay would be roughly $T/2$, where $2T$ is the width of total supply clock. In the worst case corner, SS and worst temperature 80 °C less than 1% variation in delay is observed where the time period is 10 μs. Therefore, in worst case process variation of chip and die, delay would not be affected significantly. A 3% relative delay variation has been observed with three sigma variation on the path. Though process variations do not affect the propagation delay of SAL, increasing the frequencies can minimize the propagation delay. Increasing supply voltage also enhances the speed a bit. Simultaneously, power dissipation must increase with speed. However, we should keep in mind that minimizing power dissipations would be more important than decreasing the delay in SAL.

III. Subthreshold Adiabatic Logic-Based 4-bit CLA

In this section, design and analysis of SAL-based 4-bit CLA are given to show the workability and the feasibility of the proposed logics. After verifying the logical functionality, we implemented an SAL-based standard cell library, consisting of common digital gates, such as buffer/inverter, two-input and three-input functions, complex gates, and special gates like half and full adder, which are necessary to implement the 4-bit CLA. The digital gates of the library are developed at transistor level using ramp type supply voltage as discussed in the previous section. Hence, 22-nm technology file is used in our transistor-level designs which guarantee the manufacturability of our designs under all normal conditions with favorable yields.

In Fig. 7, structures of basic logic gates considering SAL are given. These structures resemble either the pull-up or the pull-down network of the static conventional logic. For example, to implement a NAND or a NOR gate, simply the pull-up network can be placed between the supply clock and the output load capacitors, whereas an AND or an OR gate can be implemented using the pull-down network between the supply clock and the output load capacitors. In case of a NAND structure, for every input combination except $A = B = 1$, the output node voltage will follow the supply clock closely, and we get a triangular output waveform. When $A = B = 1$ through parallel pMOS transistor, leakage currents will flow as the transistors will behave almost as a constant current source. A very small amount of charge will be stored across the load capacitor, i.e., instead of ground potential, very small voltage will be dropped across the output.

The basic building block of 4-bit CLA is given in Fig. 8, which is also very similar to the conventional structure [18]. Hence, we implemented the sum (S_i) in three stages to avoid delay mismatching with the carry generation. In SAL-based 4-bit CLA, every stage will be controlled by the supply clock. Like the conventional approach, the expression of the ith sum and the $(i+1)$th carry output can be given as

$$S_i = A_i \oplus B_i \oplus C_i \tag{29}$$

$$C_{i+1} = A_i B_i + (A_i \oplus B_i) C_i. \tag{30}$$

According to the synthesized gate level block, the SAL gate level structure of 4-bit CLA has been implemented using Virtuoso(R) Schematic Composer.

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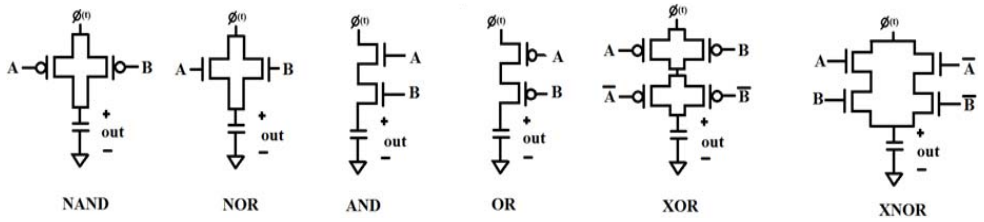


Fig. 7. Logical structure of basic SAL logic gates.

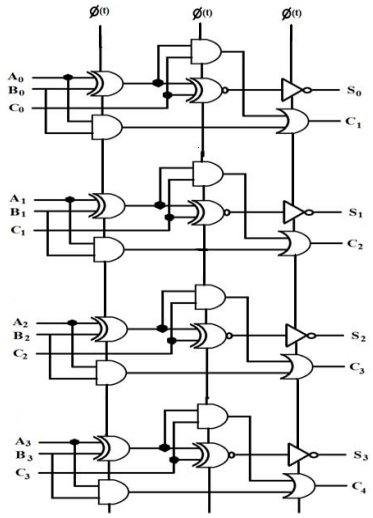


Fig. 8. Logic structure of 4-bit CLA.

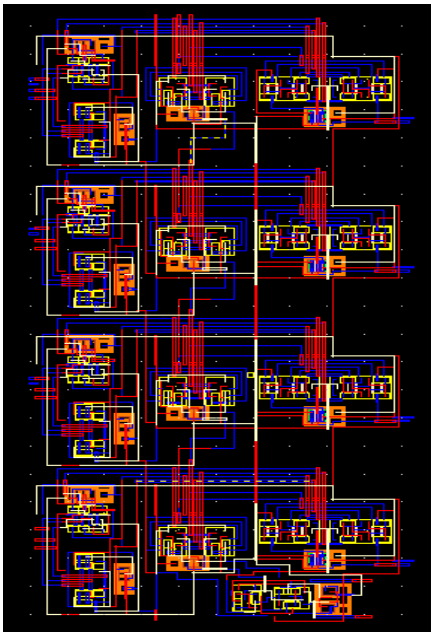


Fig. 9. Layout of 4-bit SAL-based CLA.

The layout of the 4-bit CLA is also given in Fig. 9. Special care must be taken during layout in case of routing, parasitic effect, V_{DD}, and clock rail. Design rule checking (DRC) and the circuit structure extraction are performed on the layout view of the adiabatic system. The view extracted from the layout and the original schematic views are compared with the layout versus schematic (LVS) tool simultaneously. During the physical layout design, we regularly run DRC and LVS check

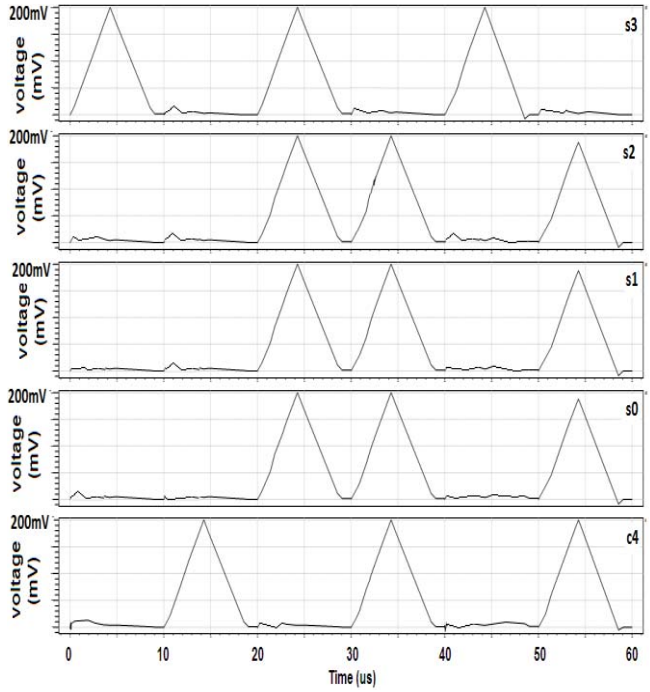


Fig. 10. Obtained output waveform of 4-bit CLA-based on SAL in postlayout simulation.

on the finished blocks. After the layout, we back annotate the layout parasitic which were unaccounted for during the electrical design and rerun the critical simulations to verify the postlayout operation of the circuit. Therefore, the design flow that we presented does not significantly differ from what the designers are used to. Hence, the transistor count will be almost half compared with the conventional CMOS logic design as in SAL basic logic gates have been implemented using either the pull-up or the pull-down transistors. Therefore, the SAL-based CLA is also area efficient in comparison with the conventional structure.

Postlayout simulations have been carried out under an analog design window to verify proper functionality. In the postlayout output waveform, given in Fig. 10, four sum outputs, i.e., S_0, S_1, S_2, and S_3 along with the final carry output, C_4, are given to show a better understanding and workability of the proposed logic. Hence, the inputs and outputs, i.e., A, B, and S are represented as, $A = A_3A_2A_1A_0$, $B = B_3B_2B_1B_0$, and $S = S_3S_2S_1S_0$, respectively. During the simulation, we set arbitraries A and B along with C_{in}. The values are $A_0 = 01000$, $A_1 = 010101$, $A_2 = 010000$, $A_3 = 010110$, $B_0 = 011001$, $B_1 = 001101$, $B_2 = 001001$, $B_3 = 101101$, and $C_{in} = 000101$. Initially, carry is set to logic 0, i.e., $C_0 = 0$. Later arbitrary logic levels are assigned

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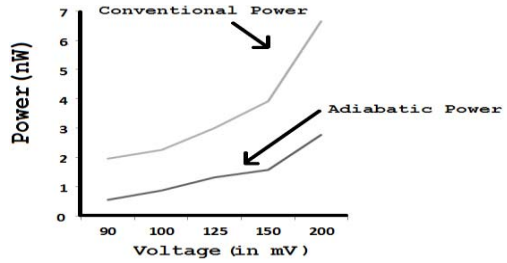


Fig. 11. Comparison of power dissipation of a 4-bit CLA between conventional static logic and SAL.

for C_{in}. For the first 10 μs, $A = (0000)_2$ and $B = (0001)_2$, which are $(0)_{10}$ and $(8)_{10}$ in the decimal system, respectively. Adding these A and B inputs, we get $(8)_{10}$ or $(1000)_2$ as outputs. Therefore, for the first 10 μs, sum bits will be $(1000)_2$. In other words, $S_0 = 0$; $S_1 = 0$; $S_2 = 0$; and $S_3 = 1$. The carry output will be $C_4 = 0$, as no carry will be generated. If carry is generated it would be forwarded to the next stage, as given in Fig. 8. Output waveforms are triangular as the charging and discharging of output nodes follow the supply clock, $\varphi(t)$, very closely.

A. Power Gain

In Fig. 11, we observe that the power dissipation of SAL-based 4-bit CLA or conventional static logic-based 4-bit CLA increases with the supply voltage. On the basis of (19) and (20), it can be said that the power dissipation due to switching depends on the square of the supply voltage, V_{DD}, in both the conventional approach and in SAL. Power dissipation due to leakage will be a function of V_{DD} also in the case of both logic, yet it can be mentioned from (24) that with increase in supply voltage, power dissipations due to leakage affect the conventional logic severely compared with SAL. Here, the conventional static CLA in subthreshold regime consumes almost 2.3–2.6 times the total power dissipated by the SAL-based counterpart for the 90–200-mV supply voltage range. Therefore, in a full cycle of $2T$ s, the power gain (=power dissipation by conventional static logic/power dissipation by SAL) will vary between 2.3 and 2.6, given in Fig. 12. As optimum supply voltage for SAL is close to 191 mV, power gain becomes maximum, i.e., 2.6 in this zone.

B. Effect of Temperature Variation in Voltage Swing and Power Dissipation

It should be noted that subthreshold logic aims at low power consumption, which is usually constrained to a few tens of Watts. In case of temperature variations, subthreshold current will vary as it depends exponentially on temperature. Therefore, output node swing can be affected due to the temperature variation. Fig. 13 shows the effect of temperature on the output voltage of SAL-based 4-bit CLA.

Here, we consider the final carry output node, C_4 node of SAL-based 4-bit CLA, and measure the voltage levels for different temperatures without changing the other parameters like supply voltage and aspect ratios. From (13), it can be mentioned that though the voltage drop across a pMOS under

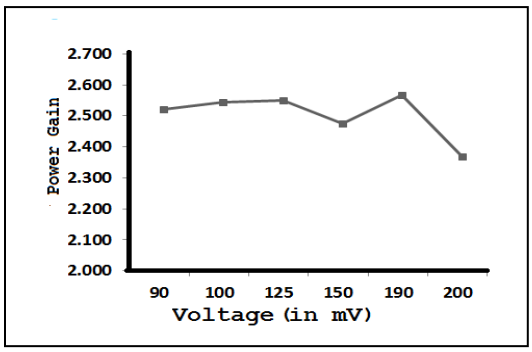


Fig. 12. Power gain of SAL-based 4-bit CLA with respect to the conventional static logic counterpart.

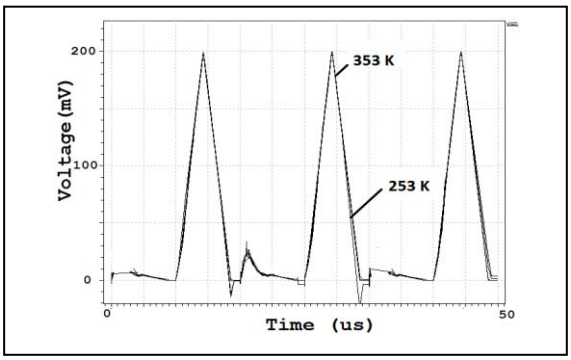


Fig. 13. Waveform of 4-bit CLA with varying temperatures.

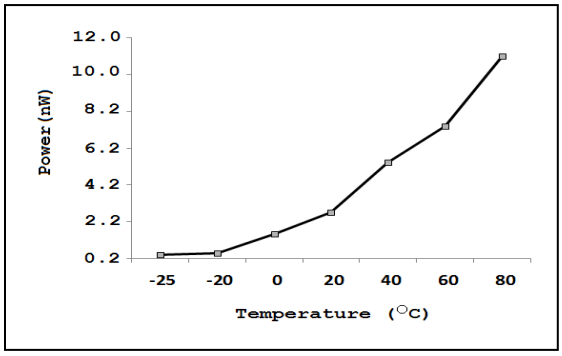


Fig. 14. Power dissipation of SAL-based 4-bit CLA for different temperatures.

low input depends on temperature, under low frequency as the width of the supply clock, i.e., T is much more, the effect of temperature variation would not affect the voltage drop so much. It has been seen that with the variation of temperature from 253 K (−20 °C) to 353 K (80 °C), the voltage degradation in the output nodes of SAL-based CLA is not much (∼5 mV). A simulated circuit can perform best in the range of 293 K (20 °C)–313 K (53 °C).

Fig. 14 shows that as we increase the temperature, power dissipation increases almost linearly from 20 °C to 80 °C. For a 50 °C temperature change, power dissipation increases more than five times compared with the power dissipation at 0 °C. Power dissipation due to switching changes depends linearly on thermal voltage $V_T (= KT/q; T = \text{temperature})$, given in (19). Therefore, switching power dissipation increases linearly with rise in temperature. With increase in temperature,

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$V_{DD}/n_P V_T$ decreases, and as a result, $\exp(-V_{DD}/n_P V_T)$ increases in (22). In addition, the power dissipation due to leakage has a direct dependency with temperature in (22). Therefore, it can be mentioned that rise in temperature would increase both the switching and leakage power dissipations. However, it should be noted that the temperature rise becomes more severe in terms of increased power dissipation in the conventional approach, detailed in (24).

Process variations are very important in subthreshold regime. Hence, ON/OFF current and variability are the main factors for subthreshold operation. In postlayout simulations for different corners are considered with varying temperature. In SS corner for hot condition, leakage power becomes more and the circuit will have the highest probability of failing. At 80 °C, under Typical-typical (TT), SS, and Fast-fast (FF) corners almost 12.9, 13.3, and 12.3-nW power dissipate, respectively. Whereas at 20 °C, under TT, SS, and FF corners almost 12.2, 12.5, and 11.8-nW power dissipate, respectively. In the worst case of SS corner with an 80 °C temperature also proper outputs can be obtained. For the worst corner, the delay certainly increased by 4% but logic degradation is absent. The variations in the atomistic nature, random doping fluctuations, and line edge roughness may affect the threshold voltage of nano-scaled devices. Fluctuation of threshold voltage would not affect the energy or power dissipation so much because the energy dissipation does not depend on threshold voltage directly, given in (26). However, under a 200-mV supply voltage and a 5-fF load in the output nodes, a 10% variation of threshold voltage introduces a 3%–5% change in the logic swing in the output nodes. Power dissipation does not vary so much (remains within 5% variation). This showed even with process variations, and we actually obtained proper waveforms and can claim that the proposed design can work commercially and would not affect the yield of the whole chip.

IV. Conclusion

SAL has been presented in this paper for the first time in the literature to advance the ultralow power research. A closed-form expression of the energy dissipation has been derived, from which insight is gained into the dependence of energy dissipation on design and process parameters. SAL saves considerable energy compared with the static conventional logic counterpart over a wide range of frequency. In particular, the impact of temperature variation on leakage dissipation, output swing, etc., has been discussed thoroughly in this paper. Hence, the predicted values of optimum frequency and optimum supply voltage almost match the simulated ones. Postlayout simulations using CADENCE SPICE Spectra and the comparison with the static counterpart explain the workability of SAL. This proposed logic scheme can be used in future energy-saving embedded circuits and mainly for energy-efficient devices where ultralow power and longevity are the pivotal issues.

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